

NexGlove

AI-Powered Wearable Human–Machine Interface

NexGlove is an AI-powered wearable technology project that aims to transform natural hand, wrist, and finger movements into intelligent, customizable interactions with digital and physical systems.

The project explores how **artificial intelligence can be combined with low-cost sensors, embedded systems, wireless communication, and software** to extract more functionality from minimal hardware.

Instead of designing NexGlove as only a gaming controller or conventional gesture glove, the long-term vision is to develop it as a **general-purpose human–machine interface** that can interact with computers, robots, IoT devices, AR/VR environments, industrial systems, and other technologies.

Core Concept



The fundamental idea is to move from simple **sensor-to-command systems** toward intelligent systems that can understand what the user is trying to accomplish.

Problem Statement

Modern interfaces rely heavily on keyboards, mice, touchscreens, buttons, and dedicated controllers. While effective, these interfaces are often application-specific and require users to adapt their behavior to predefined input mechanisms.

At the same time, increasing the number of physical sensors in wearable systems can increase cost, size, wiring, power consumption, and complexity.

NexGlove explores an alternative approach:

Can intelligent processing and AI extract more useful information from a small amount of sensor data, reducing the need for excessive hardware while creating a more natural and adaptable interface?

This approach directly supports the project's problem statement of **intelligent use of resources, technological advancement, and combining artificial intelligence with multiple sources of information to obtain valuable insights**.

Relation to the Project Problem Statement

Intelligent use of resources

Instead of continuously increasing the number of sensors, NexGlove investigates how existing sensor data can be processed more intelligently.

Technology transformation

A conventional wearable sensor system is transformed into an intelligent interaction platform.

Artificial intelligence

AI can be used for gesture recognition, movement classification, user-specific adaptation, sensor fusion, anomaly detection, and intent estimation.

Multiple information sources

NexGlove can eventually combine IMU data, button/touch input, camera/vision data, environmental sensors, device information, historical user data, and system/application context to produce a more meaningful interpretation than any individual sensor could provide.

Current Prototype

The current NexGlove prototype is based on:

- ESP32 development board
- MPU6050/MPU6500-class IMU
- Push-button inputs
- Bluetooth Low Energy communication
- BLE HID interaction
- Battery-powered operation

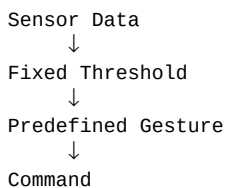
The prototype currently focuses on establishing reliable motion sensing, wireless communication, and basic human–computer interaction.

Future versions will introduce AI-based gesture recognition, adaptive control, sensor fusion, haptic feedback, and multimodal interaction.

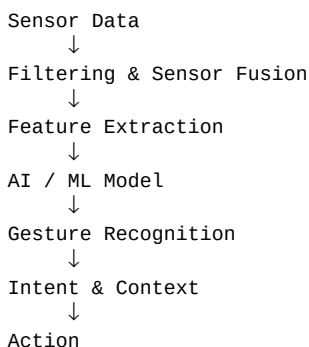
AI Integration

AI is intended to become the intelligence layer of NexGlove.

A conventional system may operate like:



NexGlove aims to evolve this into:



This allows the system to potentially learn and recognize user-specific movement patterns instead of depending entirely on fixed thresholds.

Multimodal Intelligence

A future version of NexGlove can combine multiple information sources such as:

- IMU data
- Hand and finger movement
- Camera-based vision
- Environmental sensors
- IoT device data
- Application context
- Historical data

These sources can be processed together to obtain information that would be difficult to derive from a single sensor.

```
Camera → Identifies an object
IMU → Detects user movement
IoT → Provides device information
AI → Combines the information
      ↓
User Intent / Useful Insight
```

Potential Applications

- Human–computer interaction
- Touchless computer interfaces
- Robotics and robotic-arm control
- IoT device control
- Industrial automation
- Machine interfaces
- AR/VR interaction
- Gaming and customizable controllers
- Assistive technology
- Motion analysis
- Educational and laboratory systems

The same hardware can potentially be configured for different applications through software profiles.

Customizable Gesture System

Users could eventually create their own gestures and assign actions to them.

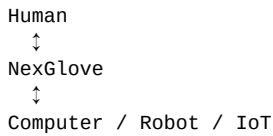
```
Gesture → Action

Pinch → Select
Swipe → Next
Rotate → Adjust
Point → Identify
Custom Gesture → User-defined command
```

The objective is to allow the user to define how NexGlove behaves rather than restricting the device to a fixed set of commands.

Haptic Feedback

Future versions can introduce haptic feedback so that NexGlove becomes a two-way interface.

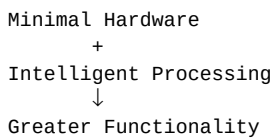


For example, a machine could send information back to the user through vibration patterns, allowing the user to receive feedback without constantly looking at a screen.

Resource-Efficient Design

A major research direction of NexGlove is **resource efficiency**.

Rather than solving every problem by adding additional sensors, the project investigates whether AI and signal processing can extract more information from existing sensors.



This can potentially reduce hardware complexity, cost, power consumption, physical size, wiring, and sensor requirements.

Development Roadmap

Phase 1 — Hardware Foundation

- ESP32 integration
- IMU communication
- Button input
- BLE communication
- Basic HID functionality

Phase 2 — Motion Processing

- Sensor calibration
- Noise filtering
- Drift reduction
- Sensor fusion
- Stable orientation estimation

Phase 3 — AI Gesture Recognition

- Dataset collection
- Feature extraction
- Gesture classification
- User-defined gestures
- Confidence estimation

Phase 4 — Adaptive Intelligence

- Personalized gesture recognition
- Adaptive sensitivity
- Context-aware commands
- False-trigger reduction

Phase 5 — Multimodal AI

- Camera integration
- Object recognition
- Sensor + vision fusion
- Intent recognition
- Context-aware interaction

Phase 6 — Advanced Wearable

- Custom PCB
- Compact hardware
- Integrated battery
- Haptic feedback
- Ergonomic wrist/finger form factor

Evaluation

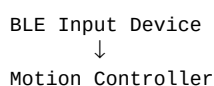
NexGlove will be evaluated using measurable engineering parameters such as:

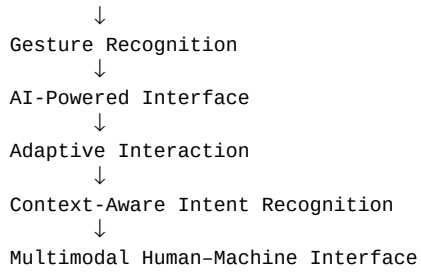
- Gesture recognition accuracy
- False-trigger rate
- Response latency
- Power consumption
- Battery life
- CPU and memory usage
- Calibration time
- Sensor requirements
- User adaptability
- System reliability

The objective is not simply to demonstrate that the device works, but to investigate **how efficiently and reliably intelligent interaction can be achieved using limited hardware resources.**

Long-Term Vision

NexGlove aims to evolve from a basic wearable input device into an **AI-powered human-machine interface** capable of understanding natural human movement and translating it into meaningful interaction with different technological environments.





Project Philosophy

Don't keep adding hardware to solve every problem. Make the existing hardware smarter.

NexGlove combines **embedded systems, sensors, signal processing, artificial intelligence, machine learning, wireless communication, and human-computer interaction** to explore a more intelligent and resource-efficient approach to wearable technology.

Tagline: From Sensors to Understanding. From Movement to Intent.